

Tables - Who Owns the Floodplain?

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Table 1. Variables and transformations

Variable	Definition & source	Unit	Transform	Notes
SFHA exposure	Parcel majority area within Zones A/AE/AO/AH	Indicator	0–1	10% overlap sensitivity checks boundary effects
Total value	Assessed parcel value	USD	Natural log	Core control
Acres	Assessor parcel size	Acres	Natural log	Core control
Improvement status	Building presence from assessor/footprints	Indicator	0–1	Used in descriptive splits & sensitivity
Owner type	Organizational form from classifier	Category	one-hot	Individual; LLC; Corporation; Trust; Gov/Nonprofit
Local portfolio footprint	SFR parcels per exact owner-of-record name (within Douglas County)	Count	Single vs multi	Exact match on owner-of-record name
Entity $\times$ footprint	Cross of owner type & footprint	Categories	one-hot	10 mutually exclusive groups; Individual—Single is the reference
Neighborhood	Assessor appraisal neighborhood	Categories	one-hot	Fixed effects; 470 groups
Owner distance	Great-circle km: parcel centroid to owner ZIP centroid	Kilometers	$\log(1+\text{km})$	Secondary robustness control

Table 2. Organizational forms: functional definitions and interpretive role

Category	Owner-of-record identification(examples)	Interpretive role in this study	Key caveats
Individual	Personal names; joint names	Household/person owner-of-record	May be a landlord; name matching may split/merge households
LLC	Tokens such as “LLC”, “L.L.C.”, “Limited Liability Company”	Business entity with limited liability; administratively legible as an entity	May be controlled by individuals, trusts, or corporations (beneficial owner not observed)
Corporation	“Inc”, “Corp”, “Co” (incorporated entities)	Business entity distinct from LLCs; may reflect corporate ownership strategies	Names can overlap with nonprofits or public entities; beneficial owner not observed
Trust	“Trust”, “Trustee” and similar markers	Fiduciary/estate-planning or investment holding structure	Not a direct proxy for rental/tenure; beneficiaries not observed
Gov/NP	Public agencies; nonprofits; religious entities	Non-market or public-interest ownership relevant for administration	Heterogeneous category; may include quasi-public entities

Table 3. Local portfolio footprint distribution by organizational form (SFR)

Entity type	Single-parcel	Single-parcel %	Multi-parcel	Multi-parcel %	Total
Individual	131,436	86.32	20,826	13.68	152,260
LLC	4,725	23.04	15,782	76.96	20,507
Corporation	762	30.36	1,748	69.64	2,510
Trust	5,295	82.54	1,120	17.46	6,415
Gov/Nonprofit	803	30.43	1,836	69.57	2,639
Total	143,021	77.59	41,312	22.41	184,333

Table 4. SFR owner-type distribution

Owner type	Count	Percent
Individual	152,260	82.60
LLC	20,507	11.12
Trust	6,415	3.48
Gov/Nonprofit	2,639	1.43
Corporation	2,510	1.36
Total	184,333	100.00

Owner type	Count	Percent
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Table 5. Model specifications

Label	Specification	Distance term	Neighborhood FE	N parcels	FE groups
M0	Entity $\times$ footprint indicators	—	No	184,333	—
M1	M0 + log(value) + log(acres)	—	No	184,333	—
M2 (primary)	M1 + neighborhood FE	—	Yes	183,313	470
M3	M1 + log(1+distance)	log(1+distance)	No	184,333	—
M4	M1 + log(1+distance) + neighborhood FE	log(1+distance)	Yes	183,313	470

Note: N differs for fixed-effects models due to neighborhood assignment and required covariates; Table 7 reports estimates for the primary model (M2).

Table 6. Counts and unadjusted SFHA rates by organizational form and local footprint

Entity & footprint	Parcels	% of SFR	SFHA count	% of SFHA	SFHA rate (%)	SFHA per 1,000
Individual — Single	131,436	71.30	1,065	52.5	0.81	8.1
Individual — Multi	20,826	11.30	260	12.8	1.25	12.5
LLC — Single	4,725	2.56	228	11.2	4.83	48.3
LLC — Multi	15,782	8.56	234	11.5	1.48	14.8
Corporation — Single	762	0.41	31	1.5	4.07	40.7
Corporation — Multi	1,748	0.95	51	2.5	2.92	29.2
Trust — Single	5,295	2.87	63	3.1	1.19	11.9
Trust — Multi	1,120	0.61	29	1.4	2.59	25.9
Gov/NP — Single	803	0.44	26	1.3	3.24	32.4
Gov/NP — Multi	1,836	1.00	40	2.0	2.18	21.8
Total	184,333	100.00	2,027	100.0	1.10	11.0

Note: SFHA counts computed from parcel counts $\times$ SFHA rates; minor rounding differences may occur.

Table 7. Adjusted risk of SFHA location by organizational form and local footprint (primary model)

Group	Risk ratio	95% CI	p-value
Individual — Single	1.000	—	—
Individual — Multi	1.209	1.038–1.407	0.015
LLC — Single	1.670	1.311–2.126	<0.001
LLC — Multi	1.379	1.104–1.722	0.005
Corporation — Single	0.989	0.626–1.563	0.963
Corporation — Multi	1.568	0.960–2.561	0.073

Group	Risk ratio	95% CI	p-value
Trust — Single	0.844	0.612–1.165	0.303
Trust — Multi	1.459	1.043–2.041	0.028
Gov/NP — Single	0.994	0.480–2.059	0.987
Gov/NP — Multi	0.792	0.348–1.799	0.577

Note: Modified Poisson GLM with log link; neighborhood fixed effects; controls for log(value) and log(acres). Standard errors clustered by neighborhood. N = 183,313.

Table 8. Within-form tests of equality between single- and multi-parcel owners (clustered)

Entity type	Difference log RR (Single – Multi)	RR Single/Multi	Wald ²	p-value	BH-FDR p-value
LLC	0.1914	1.2109	2.8347	0.0923	0.1230
Corporation	-0.4606	0.6309	4.0041	0.0454	0.0908
Trust	-0.5468	0.5788	4.6692	0.0307	0.0908
Gov/NP	0.2276	1.2556	0.8837	0.3472	0.3472

Note: p-values reflect Wald tests on coefficient differences from the primary model; false discovery rate controlled using Benjamini & Hochberg (1995).

Table 9. Residual spatial dependence diagnostic for the primary model (M2)

Moran's I (residuals)	Expected I	p-value (normal)	p-value (randomization)	k-nearest neighbors	N
0.392	-0.000005	<0.001	<0.001	8	183,313

Note: Moran's I computed on primary-model residuals using an 8-nearest-neighbor weights matrix defined on parcel centroids.